

Sarvam-M Baseline Eval Report

1. Objective

This evaluation measures the performance of Sarvam’s chat model across English, Hindi, and Tamil using four task categories: Factual Question Answering, Translation, Summarization, and Reasoning/Math. The evaluation was designed to identify strengths, errors, language-specific behavior, and response efficiency.

2. Evaluation Setup

A total of **100 prompts** were evaluated, with **25 prompts per task**:

Task	Prompts	Evaluation Method
Factual QA	25	Exact/reference-based correctness
Translation	25	chrF score
Summarization	25	Human rating: faithfulness & fluency, 1–5
Reasoning/Math	25	Final-answer exact match

The API evaluation was performed with temperature = 0.0 to make the results reproducible. The originally specified sarvam-m model was found to be deprecated/unavailable through the API, so the evaluation was performed using the currently supported sarvam-105b model. All 100/100 API requests completed successfully, and the complete outputs, latency, token counts, and errors were recorded in `results.csv`.

3. Results

Task	Prompts	Result
Factual QA	25	21/25 correct (84%)
Translation	25	14.09 average chrF
Reasoning/Math	25	23/25 correct (92%)
Summarization	25	3.68/5 correct (73.6%)

Language-level results

For Factual QA, English achieved 90%, Hindi 80%, and Tamil 80%.

For Reasoning/Math, English achieved 90%, Hindi 90%, and Tamil 100%.

For Translation, average chrF was 15.48 for English-source prompts, 5.91 for Hindi, and 19.28 for Tamil. These translation values should be interpreted only within the chrF metric and should not be directly compared with the binary QA/reasoning accuracy percentages.

4. Observed Strengths

The model demonstrated strong performance on structured reasoning and mathematical questions, achieving 23/25 correct (92%). It also performed well on factual questions, with 21/25 correct (84%). All API requests were completed successfully, indicating reliable execution of the evaluation pipeline. The outputs generally followed the requested language and provided direct answers for factual and reasoning prompts. The evaluation also demonstrates that the automated pipeline can consistently record model responses, latency, token usage, and status for every prompt.

5. Observed Failure Patterns

The factual evaluation identified several cases where the model's answer was semantically correct but did not exactly match the expected reference representation. Examples included writing "100 °C" instead of "100 degrees Celsius", using the English planet name "Mars" instead of the expected Hindi/Tamil form, or using a transliterated form such as "மார்க்ஸ்" instead of the expected Tamil reference "செவ்வாய்." These cases show that strict reference matching can classify linguistically valid answers as incorrect. Translation was substantially more difficult than factual QA and reasoning under the current chrF scoring setup, with an overall average of 14.09 chrF. The evaluation therefore indicates greater difficulty maintaining high reference-level similarity in the translation task.

6. Dataset and Evaluation Notes

The current 25 translation prompts used in the run were hand-created rather than the required FLORES-200 devtest set. Therefore, the translation results should be reported as results on the current evaluation prompt set and not described as FLORES-200 results. One translation reference (ID 26) was also corrected before the final scoring run.

Summarization outputs were successfully generated for all 25 prompts, but human faithfulness and fluency ratings have not yet been assigned. Therefore, no artificial summarization score has been included.

7. Conclusion

The completed baseline successfully evaluated 100 prompts with 100 successful API responses using sarvam-105b at temperature 0. The strongest automatically measured area was Reasoning/Math (92%), followed by Factual QA (84%). Translation produced a 14.09 average chrF on the current hand-created dataset, while summarization requires human evaluation before a final score can be reported. The main qualitative issues observed were strict reference mismatch, language/script variation, and lower translation similarity. All prompt-level results and evaluation information are traceable through the generated results.csv and scores.csv files.